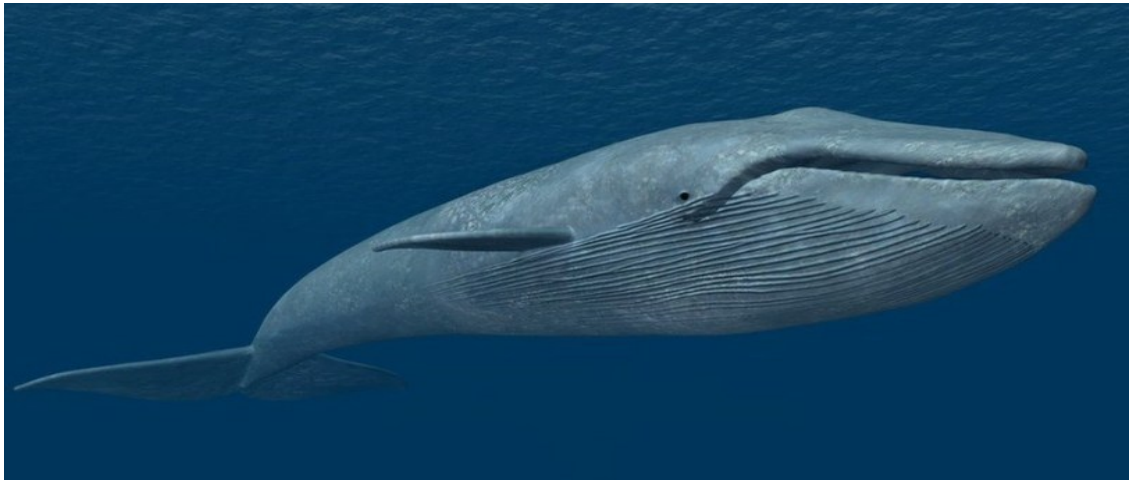
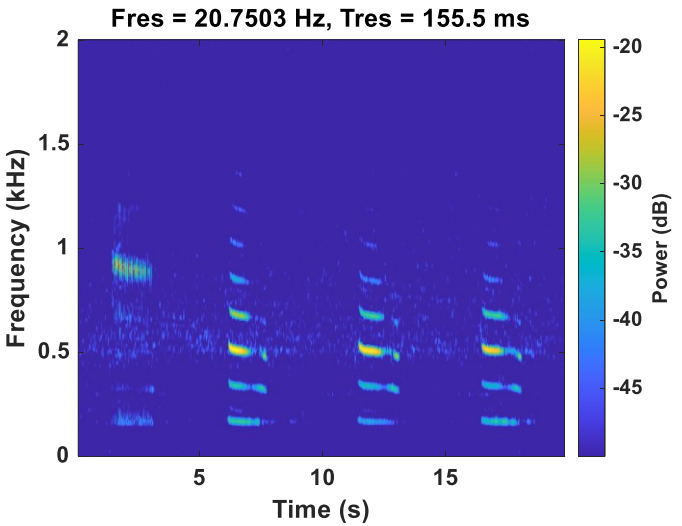
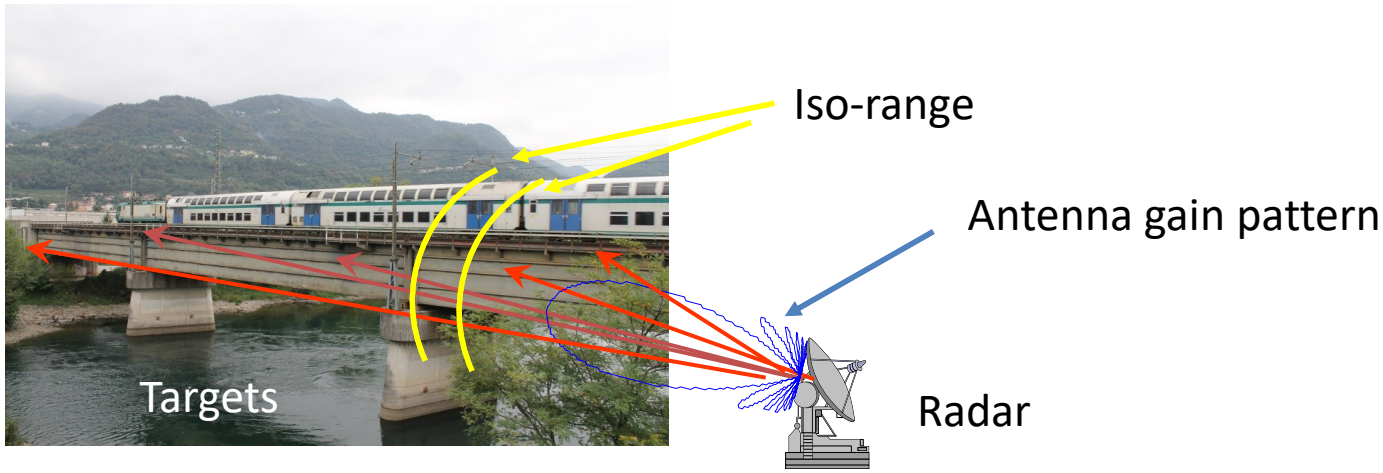
Radiometric, geometric resolution, chirps and the FMCW Radar



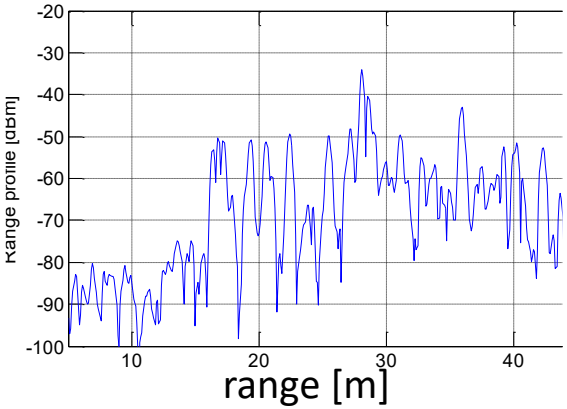


- Designing the optimal waveform
- Chirp properties
- Time-bandwidth product
- Resolution and range compression
- Transmitting a complex chirp
- Building a very simple FMCW radar

Energy, Range resolution: how to design the “best” pulse



Range profile power



$$|s_c(t, \tau_n)|^2$$

What are the properties of the optimal pulse ?

What are the feature of the optimal pulse ?

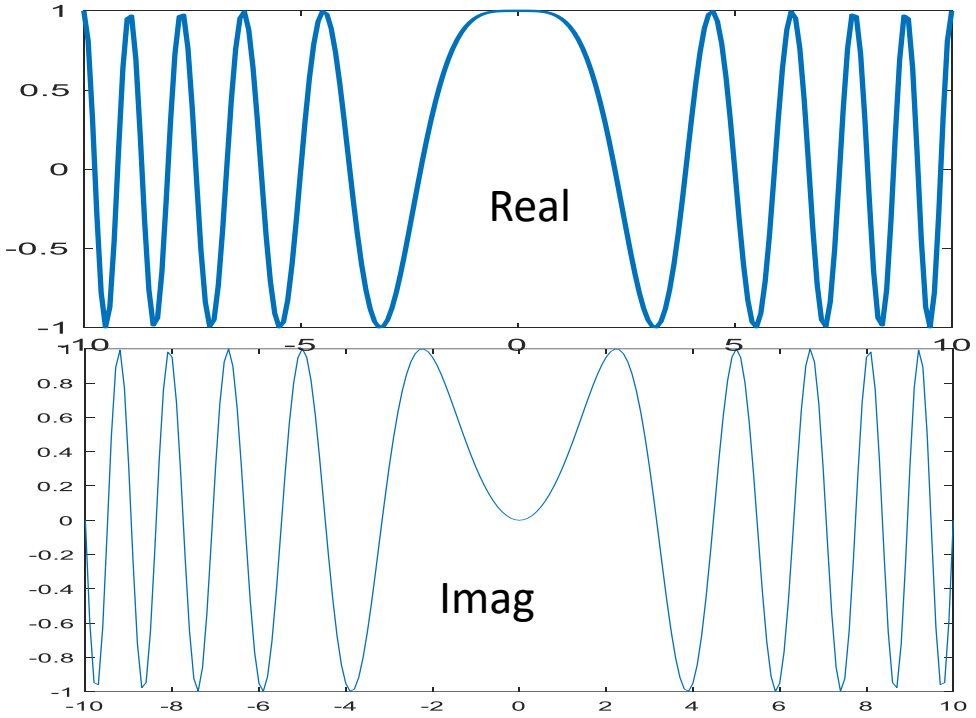
Boundary condition:

1) assume a limited bandwidth B
2) maximize SNR
3) limit the maximum peak power Pp
4) limit the pulse duration T

What is the optimal pulse:

as for SNR
and resolution (+ sidelobes) ?

The chirp



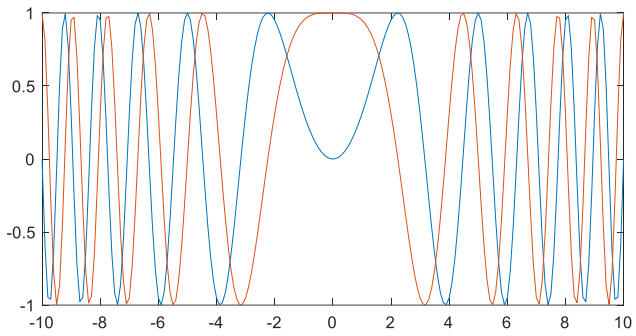
$$g(t) = e^{j\pi kt^2} rect(\frac{t}{T})$$

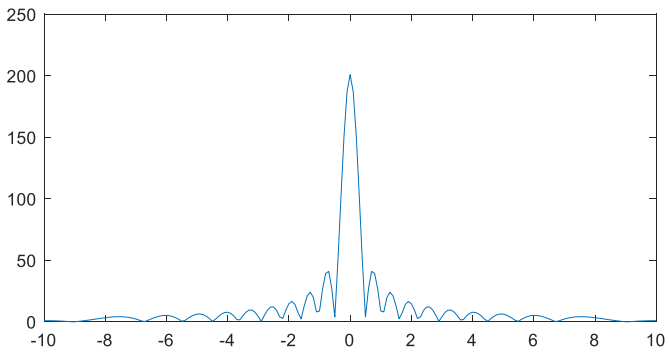
$$g_r(t) = \cos(\pi kt^2)\, rect(\frac{t}{T})$$

$$g_i(t) = \sin(\pi kt^2)\, rect(\frac{t}{T})$$

Compute the autocorrelation

The range compression



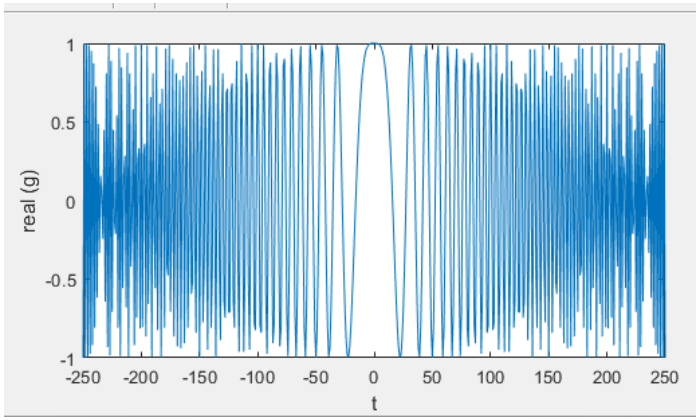


Compression factor ?

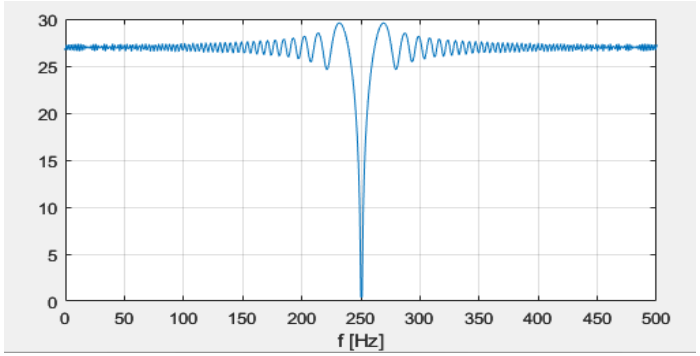
Design a complex chirp that leads to a compression factor of 500, and a time resolution of 1 second.
Write the expression of the sequence.

Hint compute: k, T, Ts=1/fs

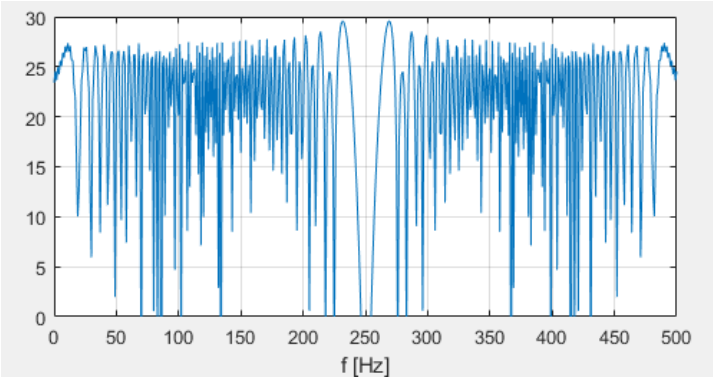
Real or complex chirp ?




```
%% Generate a complex chirp
T = 500; B = 1;
k = B/T; fs = B;
tax = -T/2:1/B:T/2;
Ch = exp(1i*pi*k*tax.^2);
Nc = length(Ch)
figure(1)
plot(tax, real(Ch))
xlabel('t'); ylabel('real (g)')
```




```
% compute and plot  FT
Nt = numel(tax);
fax = (0:Nt-1)*fs;
figure(2)
plot(fax,20*log10(abs(fft(Ch))))
grid
xlabel('f [Hz]')
ax=axis;
```




```
% same for a REAL chirp
Nt = numel(tax);
fax = (0:Nt-1)*fs;
figure(2)
plot(fax,20*log10(abs(fft(Ch))))
grid
xlabel('f [Hz]')
ax=axis;
```

1) Can you explain
the difference between the complex and the real chirp spectra ?

2) Which do you prefer

Example – sound a chirp

```
%% Make a sounding chirp
T = 10;
B = 1000;
k = B/T; fs = B;
tax = -T/2:1/B:T/2;
Ch = cos(pi*k*tax.^2);


sound(real(Ch),fs)

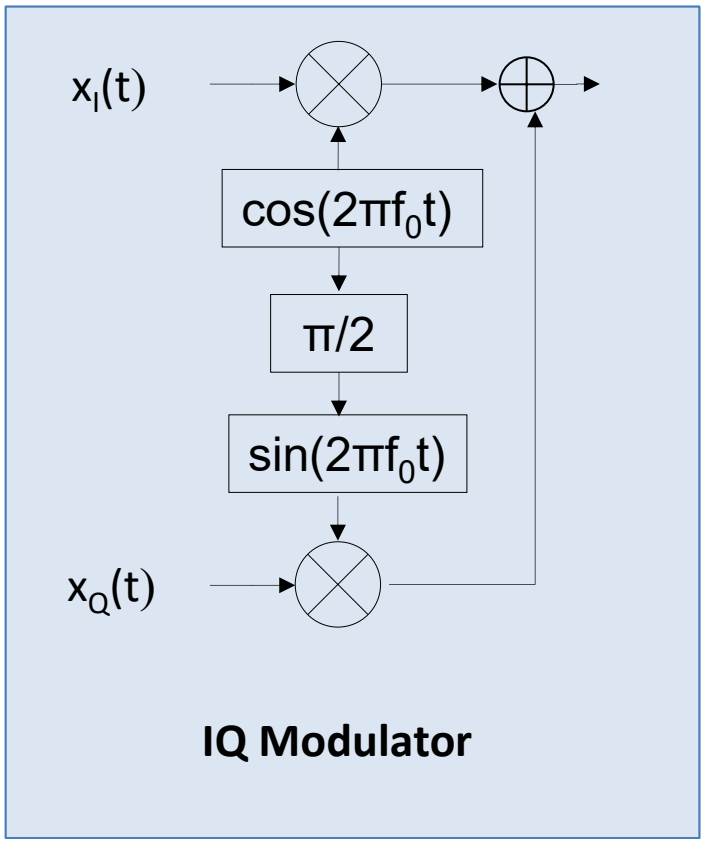

filename = 'mychirp.wav';
audiowrite(filename,real(Ch),fs);
```



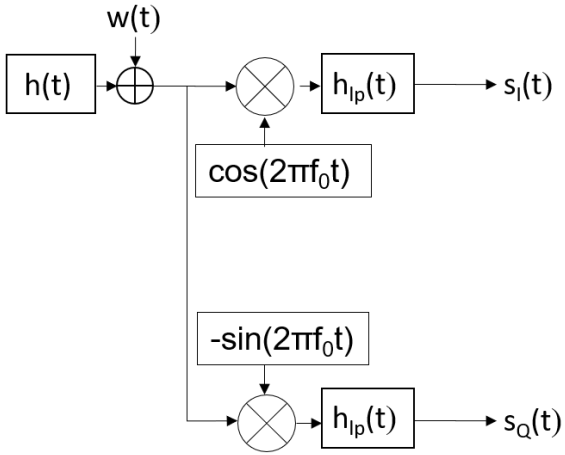
Can we transmit a complex chirp ?

If x_I and x_q are th real and the imaginary part of the chirp, compute

1) the real part of the modulated data
2) the output after medium and de-modulator

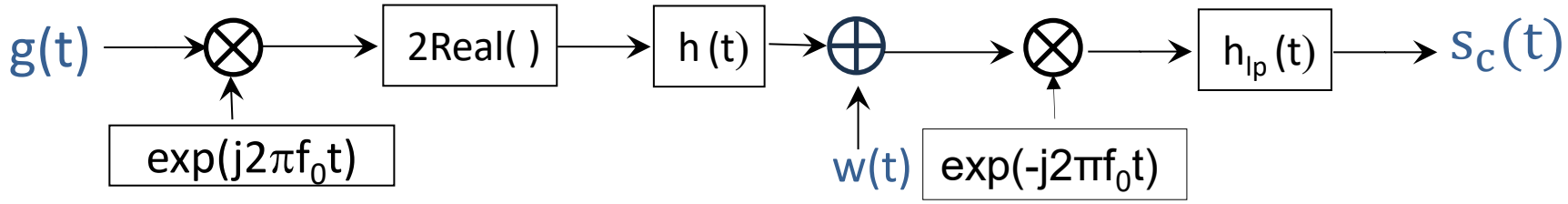


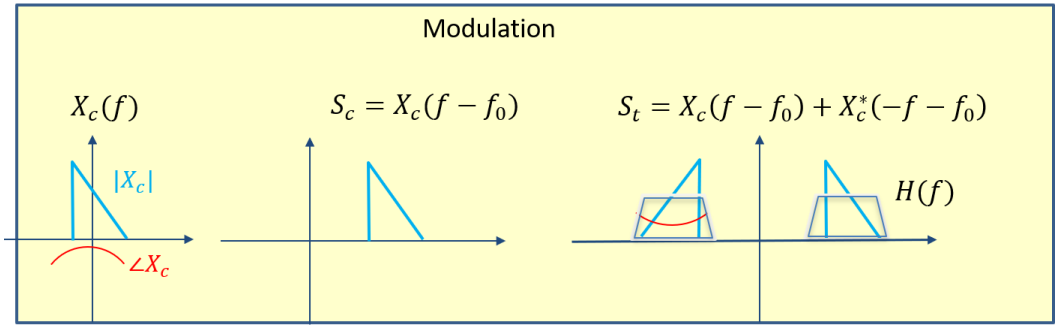
IQ Modulator

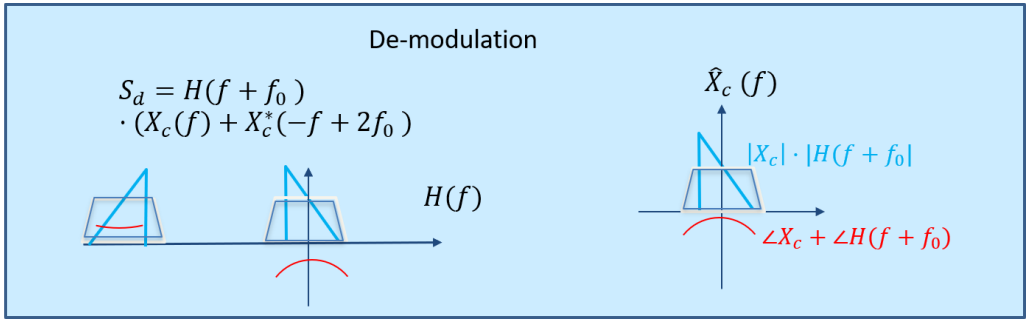


IQ De-modulator

IQ modulator + channel + demodulator

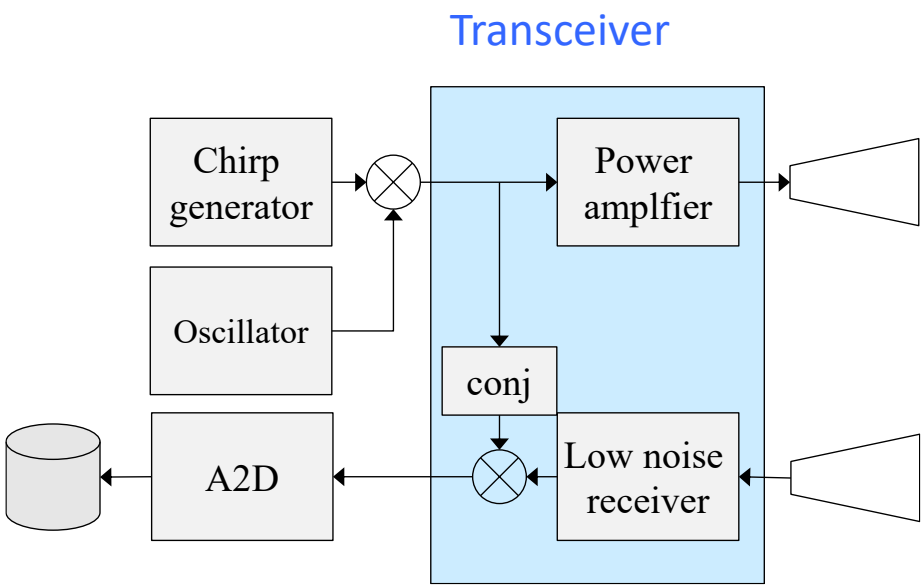






1) Associate the plotted spectra with the different steps
2) move the noise after the filter
3) Simply the scheme

Application: the Frequency Modulated Continuous Wave Radar

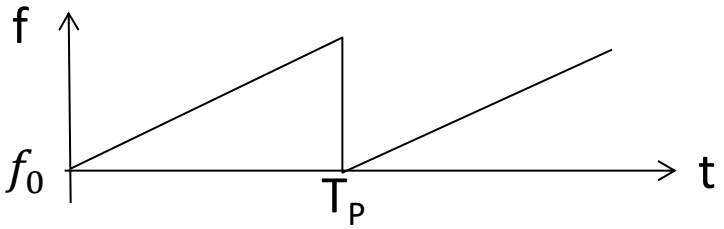




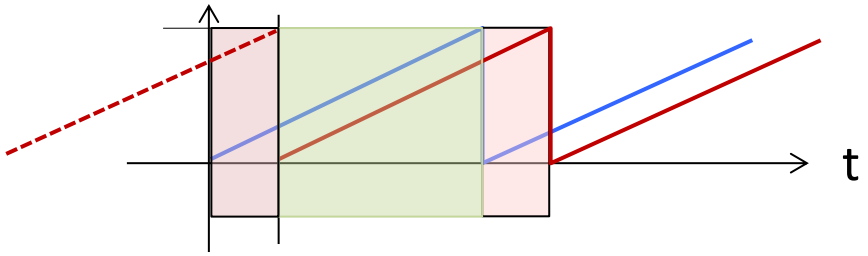
An FMCW radar transmits a continuous chirp and mixes the received echo with a coherent replica of the transmitted signal. The resulting beat signal encodes target range and velocity. A wide sweep bandwidth provides fine range resolution, which makes FMCW radar particularly suitable for automotive applications.

FMCW: timeline

Transmitted



Received (two targets, r=0 and r_{max})



1) We have two intervals: what happens in the red one ?
2) What happens if we multiply by a chirp complex conjugate, same slope
3) After that, how can we 'focus' (separate the contributions) ? Evaluate the resolution and the energy received.

Example

Design a FMCW Radar with these requirements and constraints

Item	Value	Note
Frequency	77 GHz	Constraint (automotive band)
Max range	> 5 km	Requirement from application
Max RF bandwidth	4 GHz	Constraint (ITU regulation)
Range resolution	< 10 cm	Requirement from application